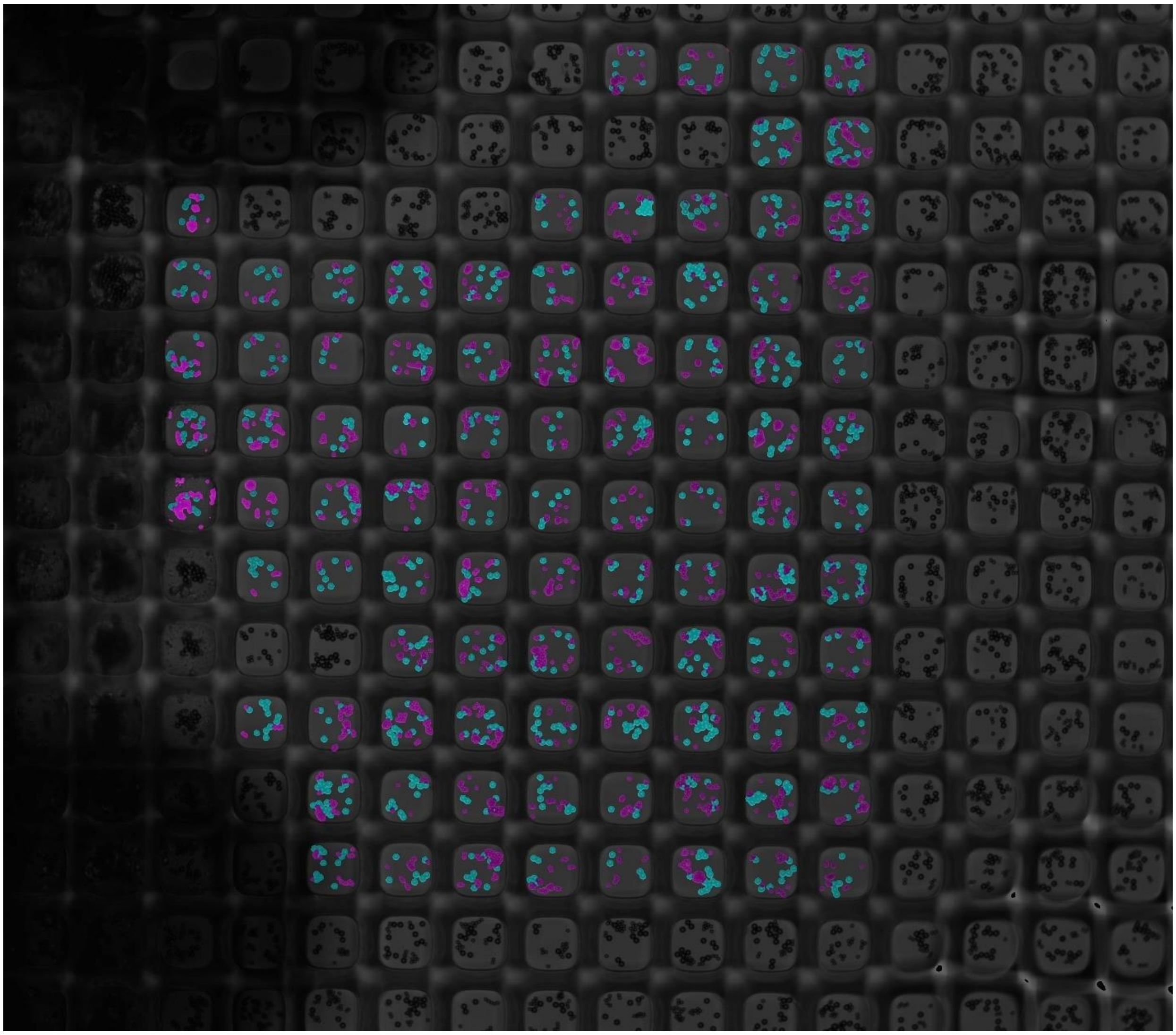

Multi-Bead ELISA Software Package

User Manual

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Introduction

The goal of this software is to segment two different bead sizes from brightfield images of nanowells and apply this segmentation to the respective fluorescent nanowell images to determine the fluorescent intensity of each bead size. Ultimately, the software will be used to measure cell secretion of multiple biomolecules through the fluorescence intensity of nearby beads in the nanowell. For example, two different sized beads, each with two different fluorescent emission ranges, can be used to determine the concentrations of up to four different biomolecules. For more information on this please refer to “Multiplexed Single Cell Secretion Assays: A Microscopy and Machine Learning Approach”.

Please note that the bead sizes are not constrained and can be chosen as long as they are visibly distinguishable in brightfield nanowell images. However, a new U-Net prediction model (Training section) will need to be trained for new sizes. This manual will primarily focus on working with 2.8um and 4.5um beads.

The software is broken into 6 sections and three scripts. The U-Net architecture script is named Model.py. Excluding the Renaming script, the 5 sections below are all integrated into a user-friendly GUI in the MultiELISAB.py script. It is recommended to download the PyCharm IDE to run this code as it makes downloading the required packages quick and simple (hovering over the imports provides immediate options to download the package). The python interpreter installed should be 3.10. Additionally, the TensorFlow version downloaded should be 2.15.0. To manually download required packages, open settings -> Python Interpreter -> + sign (install) and search the package of interest (e.g. numpy). This process should be used to download the TensorFlow package due to its specific version requirement.

Renaming: A separate python script to rename TIFF image names (microscopy images) for calibration and prediction.

Nanowell cropping: Crop nanowells and store single nanowell images.

Training and Testing: Quality inspection of training data, U-Net training, and segmentation testing.

Calibration: Applied when the concentrations of biomolecules are known. Used to determine the fluorescent intensities to create a standard curve

Prediction: Applied when the concentrations of the biomolecules are unknown. Used to determine the fluorescent intensities

Plotting: Combines data regarding nanowell locations in microscopy images and their respective fluorescent intensities. Stitches the individual nanowells with normalized intensities to display the intensity distribution in the large microscopy Tiff image.

Nanowell Cropping

This cropping functionality is developed to work with the nanowell images which have individual nanowells that are slightly smaller 160 x 160 pixels in size. The U-Net structure is also only compatible with nanowells smaller than 160x160 pixels. If larger nanowells are used, the internal code that sets the cropping size of 160x160 pixels must be changed. This change will also lead to significantly longer training times. Special thanks to Pan Deng who developed and shared the main cropping method which uses opens and closes, and a center of mass technique to center the cropping square. Figure 1 shows the nanowell slicing/cropping GUI tab.

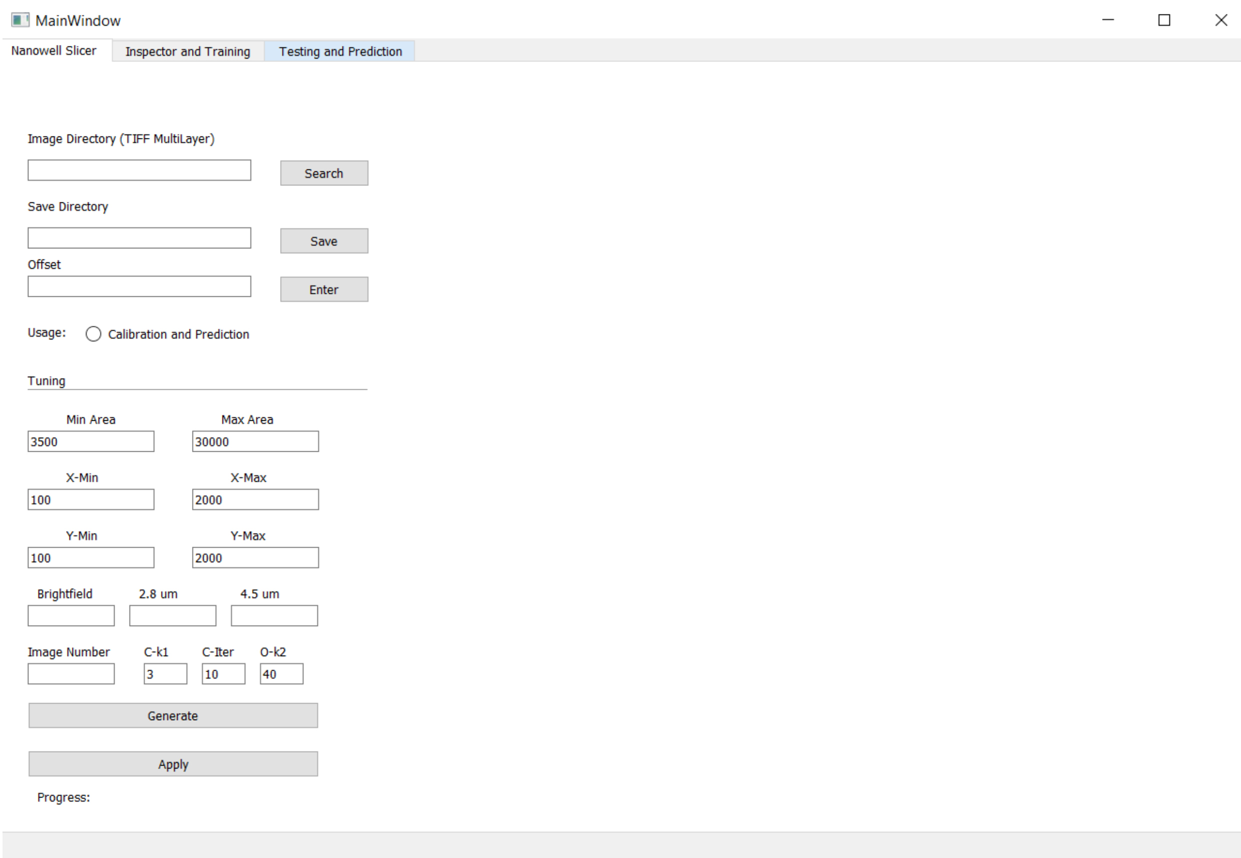


Figure 1: Nanowell Slicing/Cropping GUI Tab

In order to crop individual nanowells, the TIFF microscopy images must be placed in a single file. If the TIFF images will be used for Calibration or Prediction, please ensure that the Renamer.py script is used to rename the TIFF images. Please refer to the Renaming section regarding this. The cropping process is as follows:

1. Click the search button for the Image Directory and choose the file with the TIFFs.
2. Create a blank file or use a pre-existing file and choose this as the save directory.

3. The offset is mainly used to offset the number of the stored nanowell images in the case that there are already pre-existing cropped nanowells.
4. Calibration and Prediction should be checked when used for calibration and prediction.
5. Tuning settings:
    a. Main/Max area are used to filter incorrect detections of nanowells. Generally, the Max Area is more important. Increasing Max area can introduce incorrect cropping (partial nanowells) while decreasing can reduce the number of nanowells being cropped. A balance needs to be found.
    b. X and Y Min/Max settings are used to create a square area of the TIFF image from which nanowells can be cropped. Nanowells outside this area (e.g. on the edges) will not be cropped.
    c. Brightfield, 2.4um, and 4.5um are used to designate which images in the TIFF stack are to be used for brightfield, 2.8um bead fluorescence, and 4.5um bead fluorescence. Brightfield images are usually the first in the stack so a 1 should be inputted for them. In the case of training and testing, 2.8um and 4.5 um will be separate images in the TIFF stack since they must come from two different fluorescence's. For Calibration and Predictions, they will usually be the same image in the TIFF image stack since the fluorescence will be the same for these beads. If additional fluorescence beads are used, the cropping will just need to be repeated after changing the 2.8um and 4.5um to the next fluorescence channel in the image stack. For example, we might have green fluorescence with 2.8um used for glucagon and 4.5um used for insulin. If the fluorescence image is the third in the stack, the 3 should be inputted for both 2.8um and 4.5um and cropping should be executed. If we also have orange fluorescence 2.8um and 4.5um beads for two other biomolecules, then we simply change 2.8um and 4.5um to the location of the orange fluorescence channel in the image stack such as 4 for both 2.8um and 4.5um and crop again, preferably into a new save directory.
    d. C-k1 and C-Iter are important tuning parameters to account for brightness differences in new image acquisitions. They tune the closing operation conducted on the image for nanowell identification. Changes should be made accordingly to maximize the number of nanowells cropped correctly and minimize errors in cropping. O-k2 is a tuning parameter for the open operation in image processing but it generally should be kept near its default of 40.
    e. Finally, image number can be used to choose one of the TIFF images in the image file. For instance, input 1 to choose the first TIFF image stack in the image folder. Then click "Generate" to view the planned cropping. The cropping has still not executed yet. Continue to choose different image numbers (click generate after each time) and modify the tuning settings to maximize the number of nanowells correctly cropped. Figure 2 shows an example of the setup for nanowell cropping.

f. When the settings appear to be suitable, click “Apply” to start cropping all the TIFF image stacks in the image file. This executes the cropping for each TIFF image stack and stores the cropped nanowells in files.

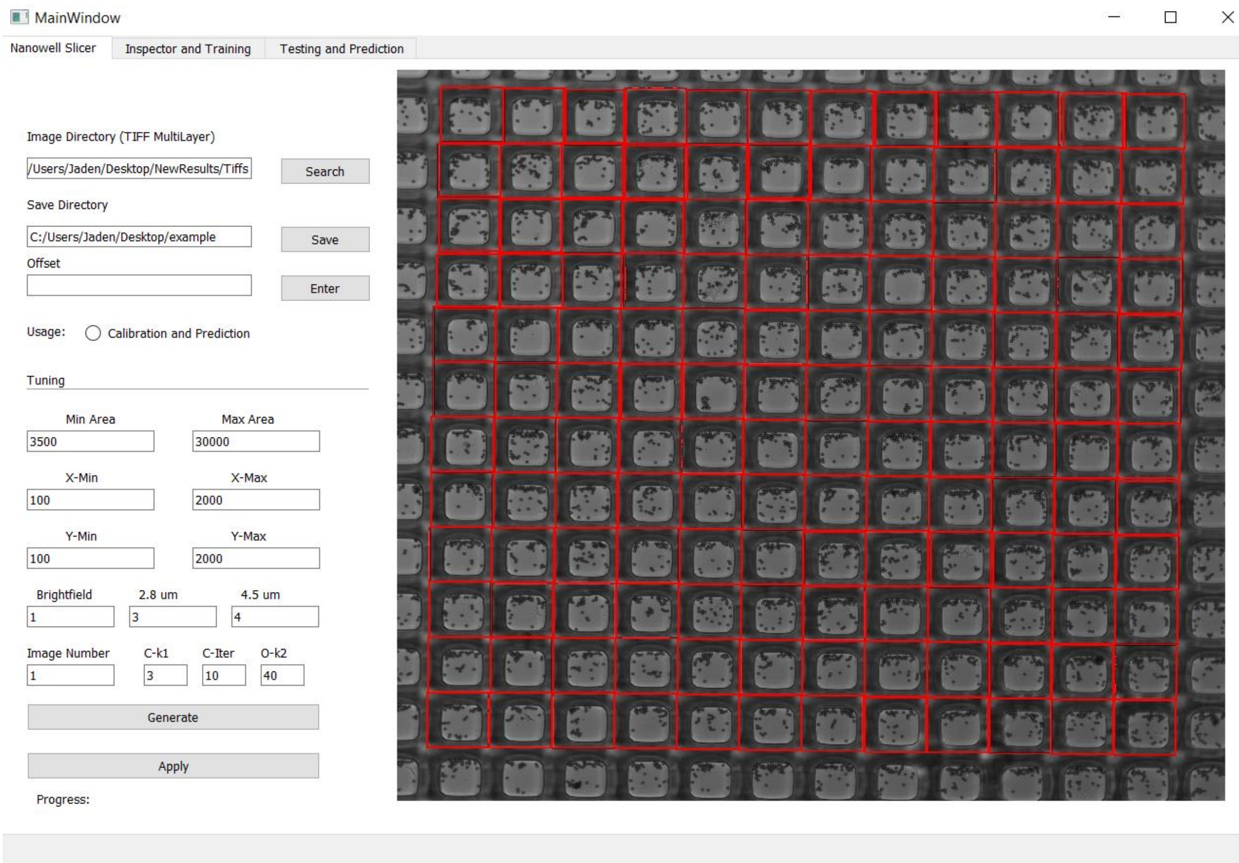


Figure 2: Example setup for nanowell cropping

The cropping program will generate files with cropped images in the save directory. Please do not modify file names if you plan to continue using this software. This software references specific file names that it creates and by changing file names, compatibility errors will occur. Additionally, a nanowell locations file is used to store the X and Y coordinates of each cropping square (center of square), the nanowell cropped image name, and the Tiff image file it belongs to. The file names created for training and testing, and calibration and prediction are shown in Table 1 and Table 2 respectively.

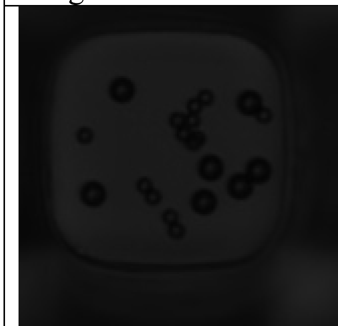
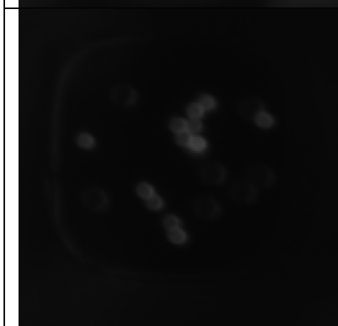
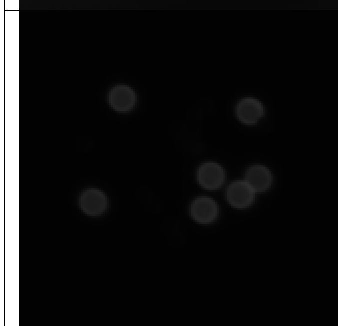
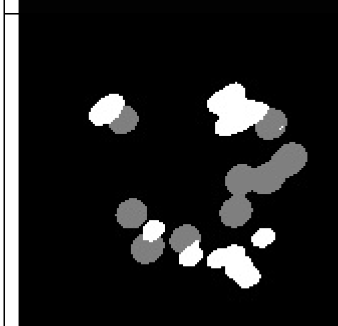
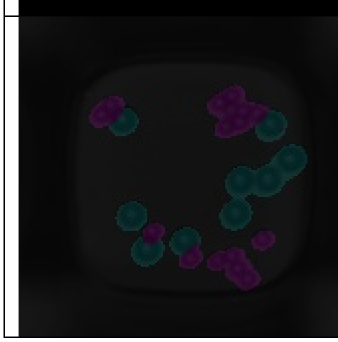
Images	Folder Name	Description
	NanoBF	Cropped brightfield images of individual nanowells
	Nano28Fl	Cropped fluorescence channel images of the 2.8um bead fluorescence
	Nano45Fl	Cropped fluorescence channel images of the 4.5um bead fluorescence
	NanoGT	Ground truth images created for each nanowell using the fluorescence images. The gray denotes 4.5um beads and white denotes 2.8um beads.
	NanoQual	Overlay of ground truths and brightfield nanowell image that is used for quality inspection.

Table 1: Folder names and images for Training and Testing Cropping

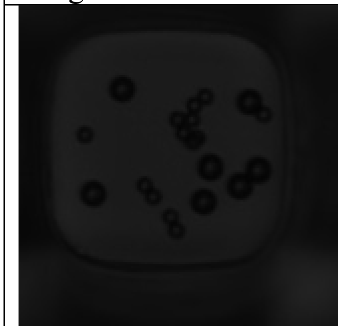
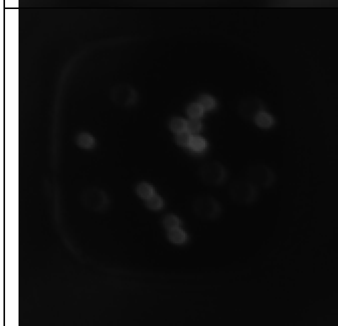
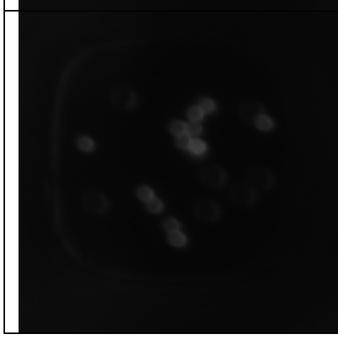
Images	Folder Name	Description
	NanoBFCalibPred	Cropped brightfield images of individual nanowells
	Nano28FlCalibPred	Cropped fluorescence channel images of the 2.8um bead fluorescence
	Nano45FlCalibPred	Cropped fluorescence channel images of the 4.5um bead fluorescence. Usually the same as the 2.8um bead fluorescence image.

Table 2: Folder names and images for Calibration and Prediction. Additional folders are created but are of no importance for calibration and prediction.

Training and Testing

Training and testing are composed of 3 sections:

1. Quality inspection and screening of training dataset
2. Dataset partitioning and Training of U-Net model
3. Testing of U-Net model

Quality inspection is conducted to ensure that the training dataset has only ground truths that correctly identify only the pixels belonging to the 2.8um bead and the pixels belonging to the 4.5um bead. Dataset partitioning is conducted to ensure enough images (5,000 – 10,000 images) are used to train the U-Net model. Finally, testing will provide dice scores for the 2.8um segmentation and 4.5um segmentations between 0 and 1 to describe the model accuracy at segmenting the different bead sizes. Dice scores above 0.7-0.8 are strong and show that the trained U-Net model can accurately determine which pixels belong to background, 2.8um beads, and 4.5um beads.

Quality inspection

Once the nanowells are cropped for training and testing, the Inspector and Training Tab should be chosen. Please ensure there are at least 3,000-15,000 cropped nanowell images to ensure that after inspection there is enough training data. The steps are as follows:

1. Select the Image directory (Save Directory from image cropping)
2. Select the save directory (using the same as image directory is recommended)
3. Auto is the sum of pixels in a nanowell ground truth. Large values usually represent nanowells with incorrect ground truths (e.g. due to background fluorescence/noise). Nanowells with ground truths that have pixel sums above the auto setting will be removed. To determine a suitable Auto setting, click the "test" check box and then click start (Figure 3). Please note the start position should generally be set to 1 unless the user wants to skip the first couple of nanowell images (e.g. they know that they are bad quality). A DataQual.csv file is created in the save directory and has the sum of pixels for each nanowell image. Using excel functions, a histogram of the data can be generated. The histogram should appear right skewed due to the higher pixel sums of incorrect ground truths. The auto can then be chosen as a value that is well to the right of the main peak (data that generally has correct ground truths) and just where the skewness is observed. Figure 4 shows an example of the plotted histogram, in this case 9000 appears to be well to the right of most data points and just at the beginning of the skewness. There is a loss of some correct ground truths with the auto function and thus it is important to choose larger values for auto and do more manual screening of the dataset to hopefully increase the training data size. Thus, the auto needs to be carefully chosen and should not be too close to the main peak.
4. Once the auto is chosen, the "test" should be unchecked, and the start button should be clicked to do manual inspection.
5. Manual Inspection requires the user to press "W" to reject the image and "E" to accept the image based on the overlayed ground truth (Figure5). The number of rejections, accepted images, total images reviewed, and images left to review are shown. If the user is unresponsive for more than 5 seconds, the program may shut down – this is completely fine, all progress is saved and the user simply must start up the program again, enter in the correct auto value and click start to continue inspection. Thus, the user does not need to inspect all images in one sitting, breaks can be taken. The files that are created after this process are shown in Table 3.

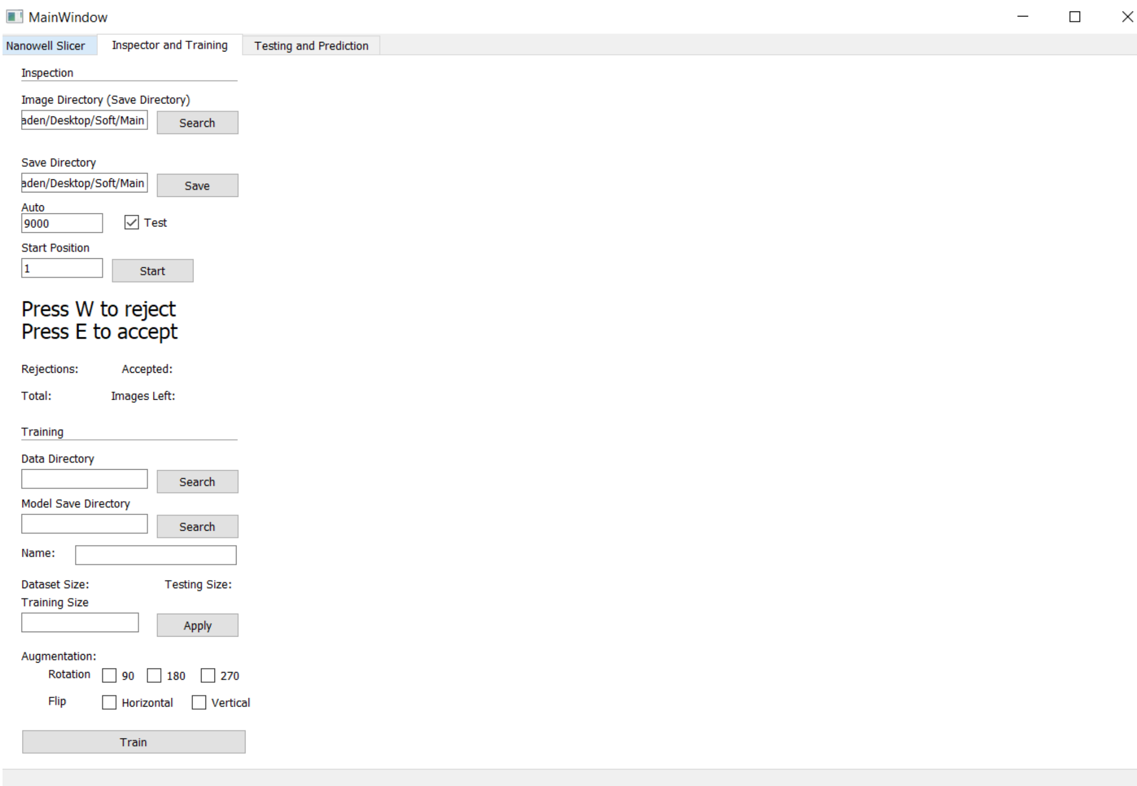


Figure 3: Setup of quality inspection for initial screening

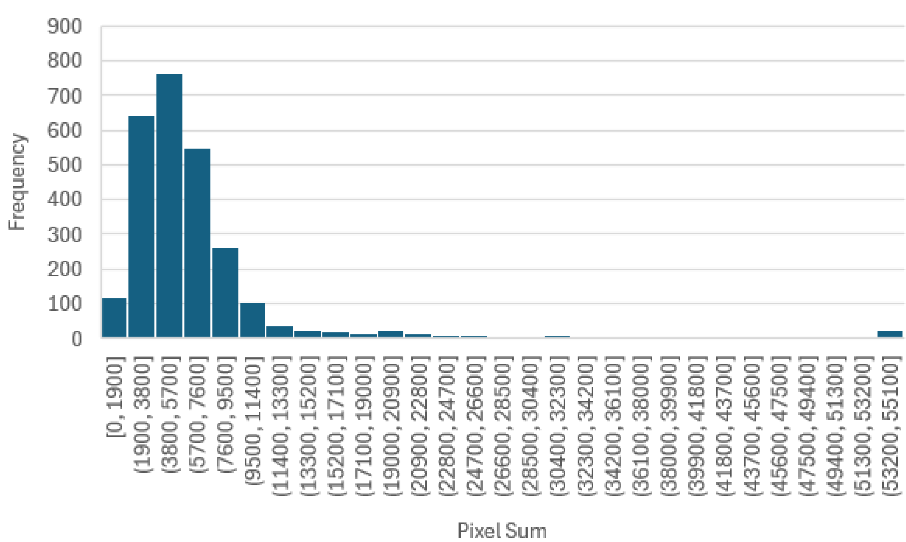


Figure 4: Histogram of ground truth pixel sum of each nanowell image.

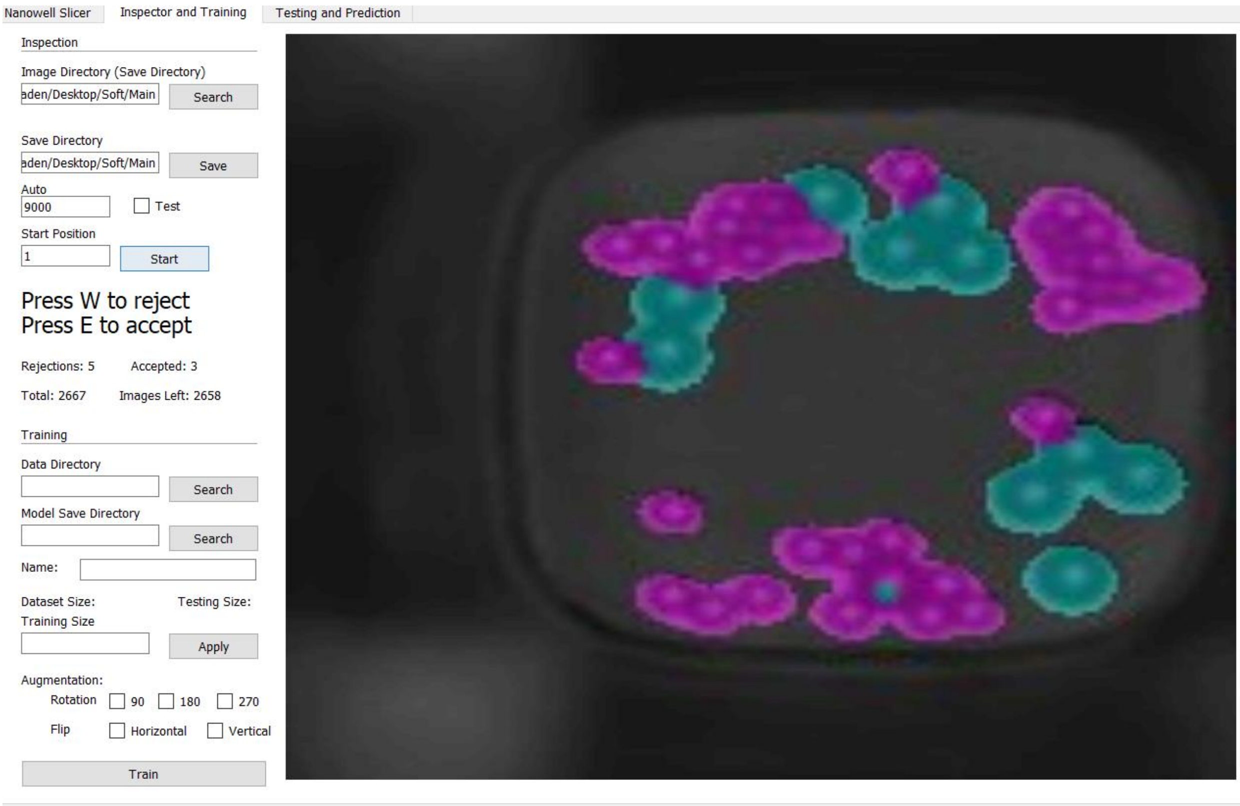
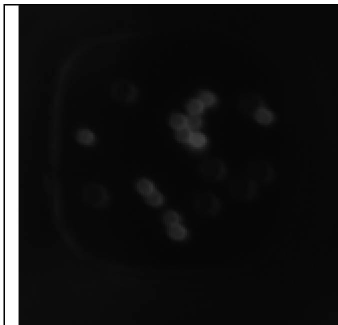
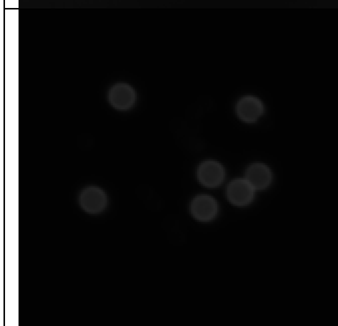
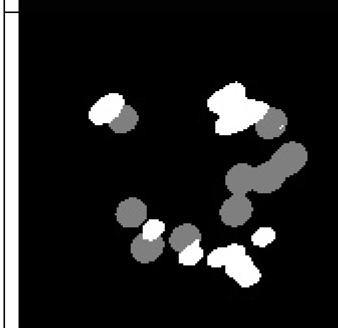
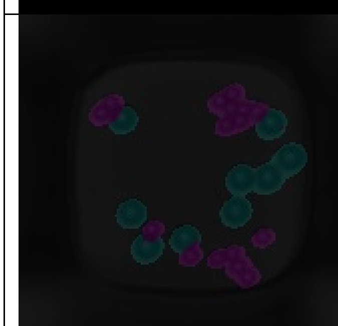


Figure 5: Manual Inspection of Training data

Images	Folder Name	Description
	goodBF	Accepted cropped brightfield images of individual nanowells

	goodQual28Fl	Accepted cropped fluorescence channel images of the 2.8um bead fluorescence
	goodQual45Fl	Accepted cropped fluorescence channel images of the 4.5um bead fluorescence
	goodGT	Accepted ground truth images created for each nanowell using the fluorescence images. The gray denotes 4.5um beads and white denotes 2.8um beads.
	goodQual	Accepted overlay of ground truths and brightfield nanowell image that was used for quality inspection.
	badQualBF	Rejected cropped brightfield images of individual nanowells

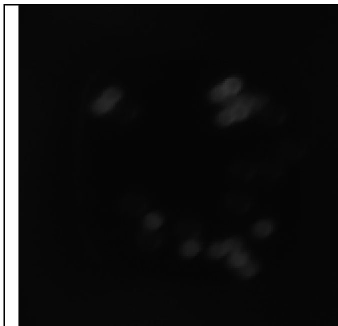
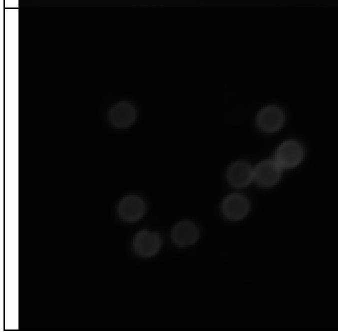
	badQual28FF	Rejected cropped fluorescence channel images of the 2.8um bead fluorescence
	badQual45FF	Rejected cropped fluorescence channel images of the 4.5um bead fluorescence

Table 3: Folder names and images after Quality Inspection

Partitioning and Training

To train the U-Net model, it is suggested that at least 5,000-10,000 training image pairs (brightfield and ground truth pairs) are provided to the model. To achieve this, the training images can be augmented by rotation or reflection across an axis to create new training data images. For instance, applying two augmentations (90-degree rotation and flip about vertical axis) will lead to a tripling of the training data set size. This makes it possible to train the model with only 1,000 – 4,000 images after the quality inspection. However, since these images are not exactly independent, it is suggested that at least 6,000 images are used when augmentations are used. Please note that before augmentation, the image dataset needs to be split up into training and testing datasets. It is suggested that the testing dataset size be about 10%-30% of the dataset size (before augmentation). The larger the testing dataset (e.g. 500-1000 testing image pairs), the more representative the testing will be. To partition the data set and train, the following steps should be taken:

1. Choose the Data Directory (usually the same as the previous save directory). This should update the GUI with the dataset size below.
2. Choose the model save directory (suggested to choose the Data Directory so it can be known what training data was used for the model)
3. Choose a name for the mode (important when training multiple times so it doesn't overwrite)
4. Then enter a training size based on the dataset size shown in the GUI. Click "Apply". This will determine the number of images for testing and will also show the dataset size.

Change the training size accordingly to ensure there are about 1,000 – 4,000 image pairs for training and about 500-1,000 image pairs for testing.

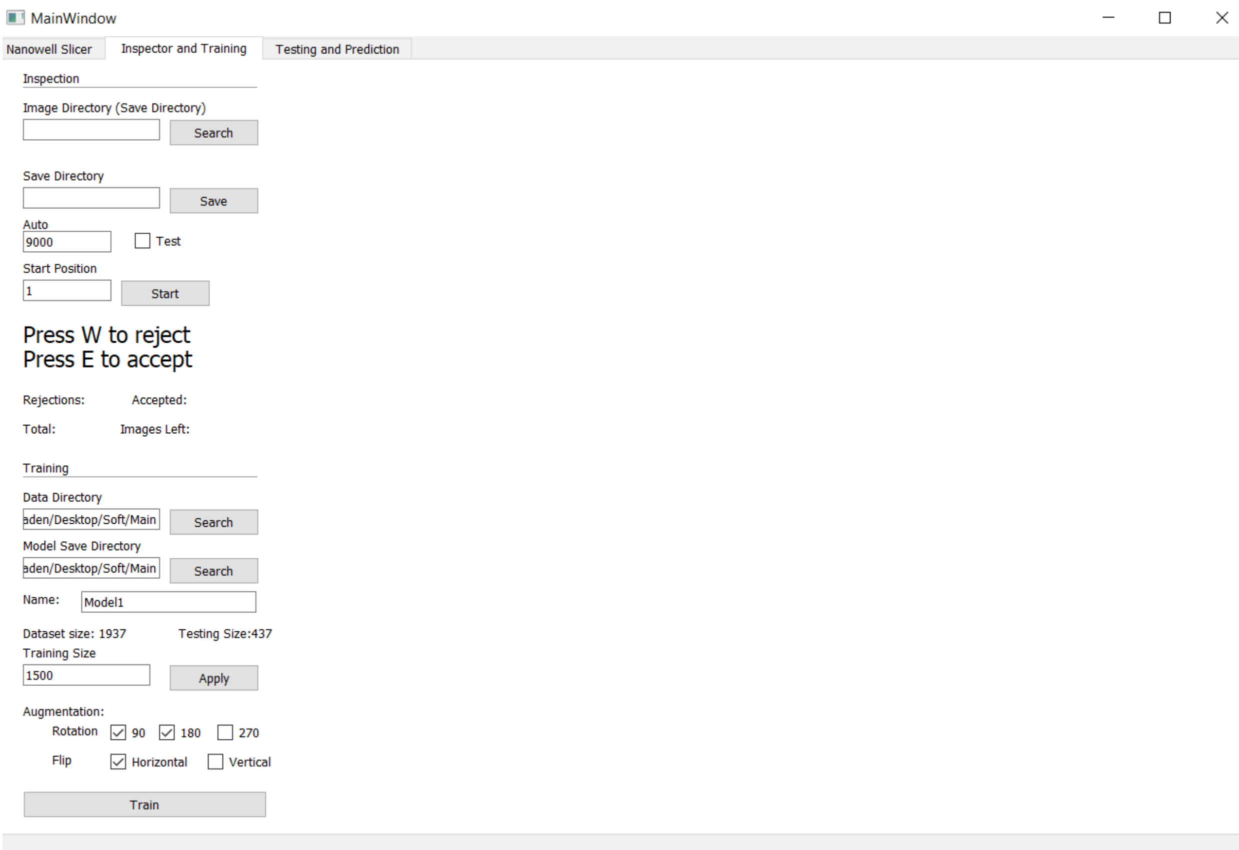


Figure 6: Setup for Training of U-Net Model

5. Select the augmentations to be conducted on the training dataset, choose enough augmentations (90 rotation, 180 rotation, 270 rotation, horizontal flip, vertical flip) to ensure the final training dataset size is greater than 6,000 images. Please note that the more augmentations chosen and the smaller the initial training dataset size is, the less independent the training data is. For the effective results, please use initial training dataset sizes of 2,000 to 4,000 and 1-3 augmentations.
6. When ready, click train and the training will start (Figure 6). Print statements will start executing in the Pycharm IDE – showing the training progress. This generally takes about 2 -3 hours. This training is not GPU accelerated since the training time is not too long and training a single model is enough as long as the training data is properly inspected. Finally, files names “testBF” and “testGT” will hold the testing image pairs along with saved numpy arrays.

Testing

The testing will provide Dice coefficients for the 2.8um and 4.5um bead segmentation. The average Dice coefficient is computed and if above 0.7, it is acceptable. In the case the average Dice coefficients are below 0.7, the boxplots outputted by the software which describe the spread of the Dice coefficients for each nanowell should be analyzed regarding the general spread. Generally, if the average is significantly below 0.7, or if there is a large spread or a group of outliers near 0, then the training data set should be increased by acquiring new training images. To test the model, click on the “Testing and Prediction” tab and follow these steps:

1. Choose the Testing Directory (usually the same as the previous save/data directory).
2. Choose the model file (model files end with a “.keras”)
3. Click “Test”. This should test the model and provide the average Dice scores with standard deviations (SD) and the minimum Dice scores, in the GUI. Additionally, it will save a plot of the boxplots of the dice scores as “DicePlot.jpg’ for the 2.8um beads and the 4.5um beads in the Testing Directory. Figure 7 and 8 shows the results of testing – Dice Coefficient statistics and Boxplots respectively. In the boxplots, 1 represents the 2.8um bead dice coefficients and 2 represents the boxplots for the 4.5um bead dice coefficients. Figures 7 and 8 show acceptable Dice Scores that indicate the model can be used for segmentation.

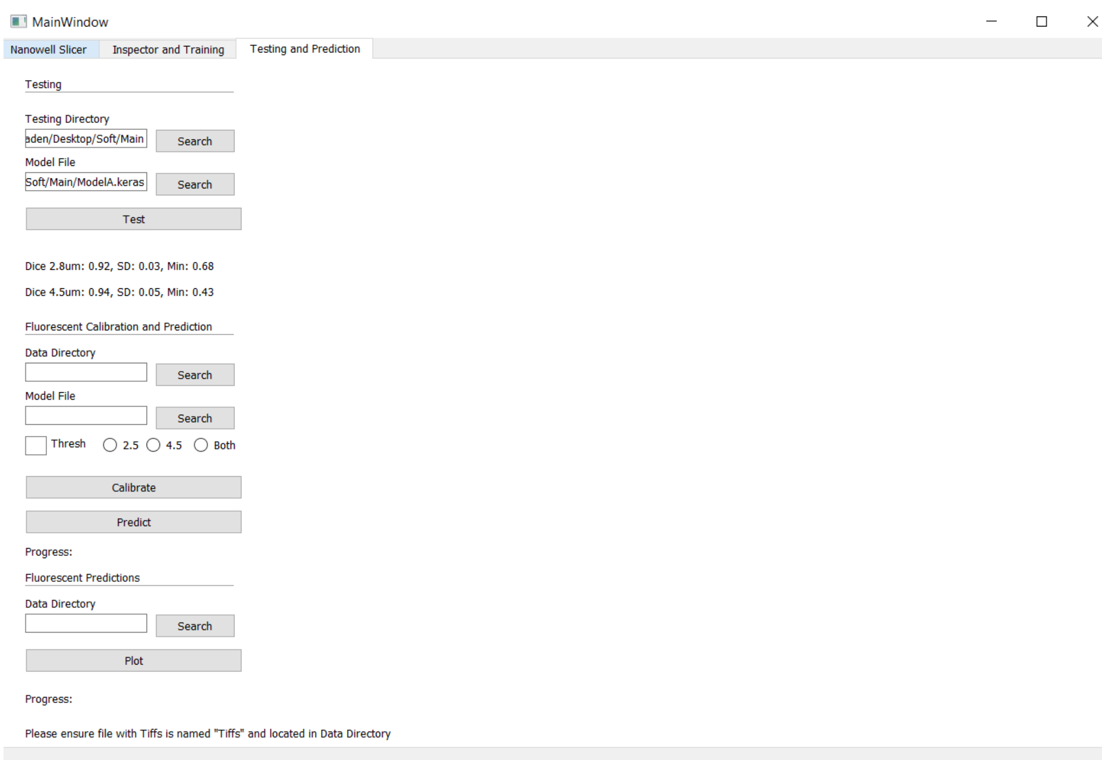


Figure 7: Results of testing shown in the GUI. The average, standard deviation (SD), and the Min Dice score are shown for the 2.8um beads and the 4.5 um beads

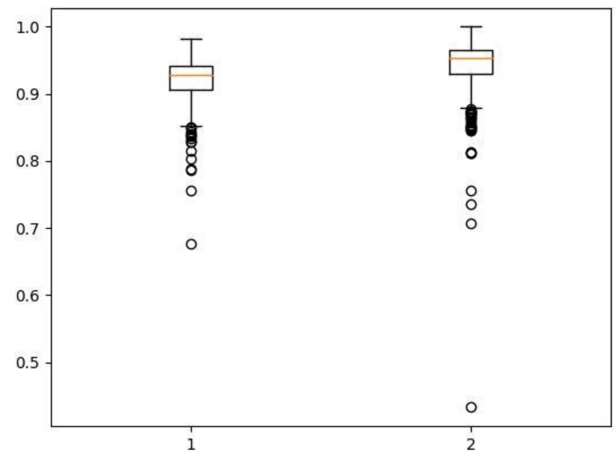


Figure 8: Boxplots of Dice coefficients for 2.8um beads (1) and 4.5um beads (2), where each data point represents a single nanowell segmentation.

Renaming

This section is specific for calibration and prediction purposes and is not needed for model training and testing. The renamer.py script will rename the TIFF images to contain the concentrations of the first and second biomolecules if they are known. If they aren't because the TIFF images are being used for prediction, then any concentration can be used to name the TIFF Images (e.g. choosing 0 and 0 for both concentrations for all images). Generally, different microwells (which contain many nanowells) contain different concentrations when creating calibration curves. Different microwells are usually labelled with different letters such as "C" or "D" in the TIFF image name. Renaming does require a very basic understanding of python coding because file names and concentrations can change greatly from time to time uses. First, we will discuss renaming for calibration and then we will discuss renaming for predictions.

To get started, the file location (file path) of the file containing the TIFF images should be set as the base_dir. All TIFF Images after renaming follow the file name convention: [Image Number - 1]_[concentration1]_[concentration2].tif. An example of the fifth TIFF image with 0 glucagon concentration and 1000 (units of choice) insulin concentration is 4_0_1000.tif. Renaming methods are completely up to user's discretion as long as the TIFF image names follow the above convention.

Calibration Renaming

The current Renamer.py script contains the “glu” array which represents the biomolecule1 concentrations for the different microwells (e.g. 1000 for “A”). The same is done for “ins” array which represents the second biomolecule. If more than two biomolecules are being calibrated for, a copy of the image file should be created, and the Renamer.py script should be re-run on the copy with the new concentrations for biomolecules 3 and 4.

It is possible to calibrate multiple concentrations at the same time. However, it is suggested to calibrate a single biomolecule at a time without any other biomolecules present. In this way we can ensure that there are no interactive effects between the biomolecule being calibrated and any other biomolecules. In future, interaction studies can be conducted to create a plot of fluorescence intensity as a function of the concentrations of two biomolecules. This will show if there are interactive effects between biomolecules that need to be accounted for.

Prediction Renaming

When using TIFF images for prediction, the concentrations don’t matter and the same calibration renaming script can be run. However, it might be easier to keep both concentrations at 0 so it is more distinctive. The concentrations chosen are up to the user’s discretion and once the images are inputted for predictions, the concentrations are removed from the results, thus holding no importance other than holding the naming convention for prediction to occur.

Calibration

The main purpose of calibration is to create a standard curve that relates fluorescent intensity and biomolecule concentration. This standard curve can then be coupled with predicted fluorescent intensities of beads in a secretion assay to determine the concentration of the biomolecule secreted. Microwells are given different biomolecule concentrations which result in different fluorescence intensities due to these different concentrations. Single biomolecule calibration will be defined as the calibration of a single biomolecule using different concentrations in different microwells and no other biomolecule present. Multi Biomolecule calibration is used to calibrate with two biomolecules present at different concentrations – its main purpose is to determine if the biomolecules interact. Interaction means that biomolecule concentrations do have independent effects on fluorescent intensity. If they are not independent, then techniques of biomolecule detection should be re-examined to ensure that there is no cross binding occurring with the different detection beads.

Single Biomolecule

For single biomolecule calibration, please ensure that if there are other biomolecules present, their concentrations are constant and there are no cross binding/interaction effects. The steps are as follows:

1. Follow the Calibration Renaming section to ensure correct file names are created for each TIFF image stack
2. Follow the Nanowell Cropping section and this time check the “Calibration and Prediction” option
3. Choose the Data Directory (this should be the save directory in cropping)
4. Choose the Model File (file name ends with “.keras”)
5. Enter a “thresh” which is pixel value threshold. Pixel values below this threshold will not be accounted for in the fluorescent images. This is used to generate fluorescent measurements that are much more like Nikon Instruments Software which uses bright spot detection for the fluorescent measurements. Bright spot detection generally measures the fluorescent circles with high pixel intensities – thus not accounting for all fluorescence that belongs to the bead. Comparatively this model uses a mask to measure the fluorescent pixels which pertain to each bead size and thus measures lower intensity pixels too. Using a proper threshold value, the standard curves from the two software will be much more similar. Please note, adjust this wisely since calibration time takes about 2-3 minutes for every 1000 nanowell images. Also, please take note of the thresh value as this must be used when predictions are done, otherwise the standard curve can’t be used to translate fluorescent predictions into concentrations.
6. Choose either “2.8”, “4.5”, or “both”. “2.8” or “4.5” should be chosen if the only the 2.8um beads or 4.5um beads are present. By choosing one of these options, the data will filter out any incorrect measurements. For instance, when the bead size that is not being used was determined to have a fluorescent intensity. Choose the “both” option when both bead sizes are present. The completed calibration section is shown in figure 9.

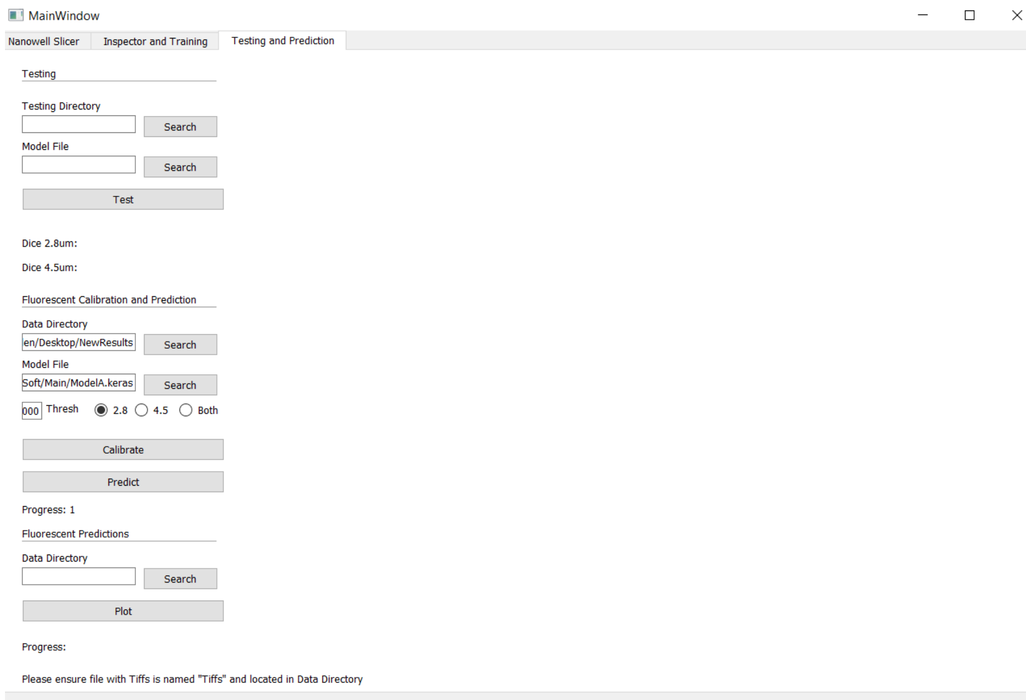


Figure 9: Completed calibration section

7. Click the “Calibrate” button and once complete, two files will be provided in the Data Directory: FluoroResults.csv and FluoroAvgResults.csv. The first file contains the biomolecule concentrations and the fluorescent measurements for each nanowell image. The second file contains the average and standard deviation of fluorescent measurements for each of the biomolecule concentrations. If the user would like to see the averages based on the TIFF image, the user must change the renaming script to rename each TIFF image stack as different concentrations. Example images of the results are shown in figures 10 and 11. All mask predictions are stored in the data directory under the folder: predictedMaskCalibPred.
8. Use external software such as MATLAB or R to perform non-linear regression on the data to create a standard curve.

Name	Glu	Ins	2.8Int	4.5Int
0_1000_0.j	0	1000	11142.04	0
0_1000_1.j	0	1000	10860.38	0
0_1000_10	0	1000	11292.22	0
0_1000_10	0	1000	13187.61	0
0_1000_10	0	1000	7147.111	0
0_1000_10	0	1000	8798.068	0
0_1000_10	0	1000	9090.904	0
0_1000_10	0	1000	10291.02	0
0_1000_10	0	1000	8452.761	0
0_1000_10	0	1000	7909.155	0
0_1000_10	0	1000	8313.755	0
0_1000_10	0	1000	12463.47	0
0_1000_10	0	1000	12355.54	0
0_1000_10	0	1000	12273	0
0_1000_10	0	1000	12544.37	0
0_1000_10	0	1000	12168.41	0
0_1000_10	0	1000	13063.96	0
0_1000_10	0	1000	12187.48	0
0_1000_10	0	1000	11681.88	0
0_1000_10	0	1000	11095.59	0
0_1000_10	0	1000	10642.08	0

Figure 10: Results from the FluoroResults.csv. Glu and Ins represent biomolecules 1 and 2.

Glu	Ins	2.8Int	2.8STD	4.5Int	4.5STD
0	1000	10452.27	2250.929	0	0
0	500	11504.09	1441.79	0	0

Figure 11: Results from the FluoroAvgResults.csv. Int represents the average columns and STD represents the standard deviation columns. Glu and Ins represent biomolecules 1 and 2.

Multi Biomolecules

Two Biomolecules

The software enables the user to also calibrate more than two biomolecules at the same time. Since interactive effects have not been examined yet, these should be investigated. When there are interactive effects, the standard curve loses its meaning – interaction between biomolecule concentrations results in a change in the fluorescent intensity that is not expected due to an interaction term being introduced. This can result in fluorescent intensities corresponding to multiple biomolecule concentrations. Thus, it is crucial that there is no interaction between biomolecules and that their fluorescent intensities are independent.

To check independence, the same process as single biomolecule calibration must be repeated. However, this time the acquired data used must have biomolecules concentration pairs that change with each other. An example of this would be a square grid of values for each biomolecule's concentration such as (1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3). 1,2,3 represent three different increasing concentrations of (biomolecule 1, biomolecule 2), and each pair represents the concentrations provided to a single microwell (a total of 9 microwells are needed in this case). Finally, the "both" option should be chosen in the calibration menu to ensure fluorescent intensity values are measured for both. Once the results are outputted into spreadsheets, the fluorescence values for each nanowell at each concentration can be plotted in 3D and a multiple linear regression analysis can be conducted to determine if the two biomolecules are independent regarding their standard curve. Two multiple linear regressions are to be performed since fluorescence values are measured for biomolecule 1 and for biomolecule 2. Software such as R can perform this analysis and will determine if there is meaningful interaction (dependance). If there is dependance, then the lab fluorescence/detection reagents must be planned again carefully to ensure that there is no cross-binding and thus no interaction effects. Once there is no interaction, the two standard curves can then be created from the results, as done for the single biomolecule case.

Three or More Biomolecules

For cases regarding more than two biomolecules, please follow the calibration renaming section to ensure that for every two additional biomolecules, a copy of the images with concentrations of the next two biomolecules are renamed. These renamed TIFF images should be cropped with the same settings previously used but with the respective fluorescence images of the next two biomolecules (cropping section), stored in a new folder to ensure calibration results are not overwritten, and checked for independence. Additionally, the pandas module in python can be used to combine the calibration results from each of the biomolecule pair result CSVs based on the image "Name" column to create a single results CSV with all biomolecules. Multiple Linear Regression tests should be run for each possible biomolecule pair combinations possible and for each biomolecule fluorescence measurements.

For instance, if four biomolecules are used with two different concentrations for each to conduct interaction testing, at least 11 microwells are needed. This is because each microwell corresponds to a specific set of concentrations and interaction testing is done in pairs. Thus with 4 biomolecules, there are 6 ways to pair the biomolecules (4 choose 2), 4 ways to change only one biomolecule concentration, and 1 way where all biomolecule concentrations are the same (unchanged). Thus, a total of 11 experimental groups are required. The formula for the number of experimental groups when two different concentrations are used, is shown below. B is the number of different biomolecules.

$$Experimental\ Groups = \sum_{n=0}^{2} \binom{B}{n}$$

The introduction of more concentration levels (e.g. 3 concentrations) leads to much larger experimental group sizes and a much more complex calculation. Through a brute force method, it can be shown that when 3 concentrations are used with 3 different biomolecules, at least 19 experimental groups are needed to conduct all possible pairwise multiple linear regressions. Therefore, careful consideration is required when planning interaction experiments.

Example

Going back to the example of 4 biomolecules and 2 different concentrations for each, we see that two files will be created for the cropping of each fluorescence image (two biomolecules per file). The renaming script will be run twice with the biomolecule concentrations being modified for each run. The 2.8um and 4.5um settings in the cropping section should be changed for each file to capture the correct fluorescence image in the TIFF stack. For example, biomolecules 1 and 2 have the same fluorescence and are in the third image in stack so 3 should be entered into 2.8um and 4.5um fields. Biomolecules 3 and 4 have the same fluorescence and are in the fourth image in the stack so 4 should be entered into the 2.8um and 4.5um fields during the second time cropping.

The multiple biomolecule calibration will be run for both files and the results should be combined by writing a python script and using the inner-merge function with pandas and the "Name" columns of the result CSVs as the requirement for merging. There are now four different biomolecules and four different fluorescence measurements for each set of concentrations used (11 in this case). There are 6 pairings (4 choose 2) for the 4 biomolecules and thus there are 4 fluorescent measurements (fluorescent measurement for each biomolecule). Therefore, a total of 24 multiple linear regressions should be conducted, one for each biomolecule combination and fluorescent measurement for each pair. Once there are no interactions found, single biomolecule concentrations can be changed while the others are held constant to gather enough fluorescence vs. biomolecule concentration data to create single biomolecule standard curves. For instance, if 5 different concentrations are chosen, then 5 concentrations x 4 biomolecules = 20 microwells are required to gather the required data for the 4 standard curves. Please refer to the single biomolecule section when creating standard curves.

Prediction

The main purpose of prediction is to measure the fluorescent intensities of the different bead sizes in each nanowell and apply the standard curve from calibration to translate the fluorescent intensities into biomolecule concentrations. When a cell is present in the nanowell, this becomes a secretion assay and when multiple biomolecules are being detected (different bead sizes or fluorescence), this becomes a multiplexed secretion assay. The process is the exact same as calibration except that the same "thresh" setting used to create the standard curve must be used when predicting. The prediction process is repeated for every 2 biomolecules, so for a prediction of 4 different biomolecules, the process must be conducted twice.

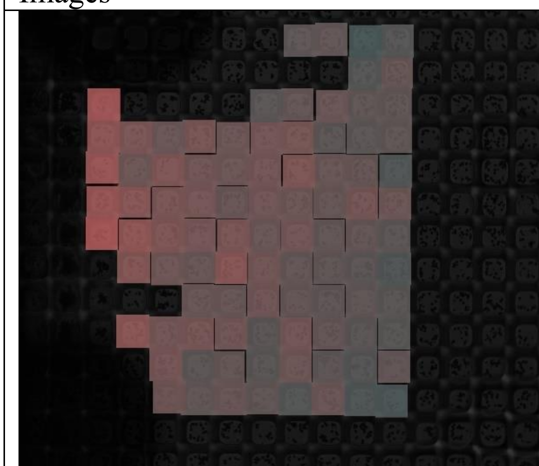
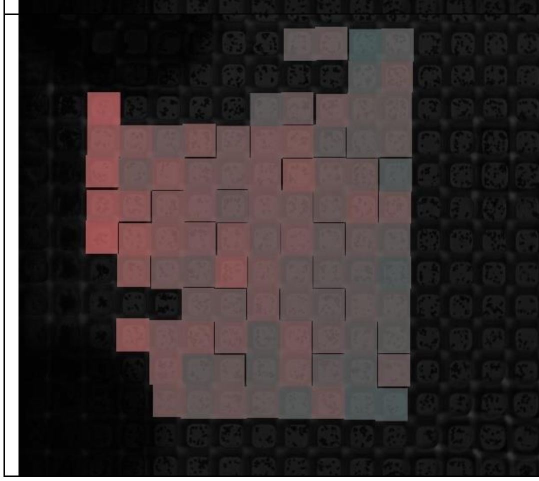
1. Follow the Prediction Renaming section to ensure correct file names are created for each TIFF image stack
2. Follow the Nanowell Cropping section and this time check the “Calibration and Prediction” option. If the biomolecules have the same fluorescence being detected (different bead sizes), then enter the same image number in TIFF image stack for both “2.8um” and “4.5”um fields in the cropping section.
3. Choose the Data Directory (this should be the save directory in cropping)
4. Choose the Model File (file name ends with “.keras”)
5. Enter the “thresh” value that was previously used to create the standard curve that will be used in this experiment to translate the fluorescent intensities into biomolecule concentrations.
6. “2.8”, “4.5”, or “both” play no role in prediction. Click the “Predict” button and once complete, the results file will be provided in the Data Directory as FluoroResults.csv. An example of the results is shown in figure 12. The results should be coupled with the standard curve to translate the fluorescent intensities into biomolecule concentrations. All mask predictions are stored in the data directory under the folder: predictedMaskCalibPred.

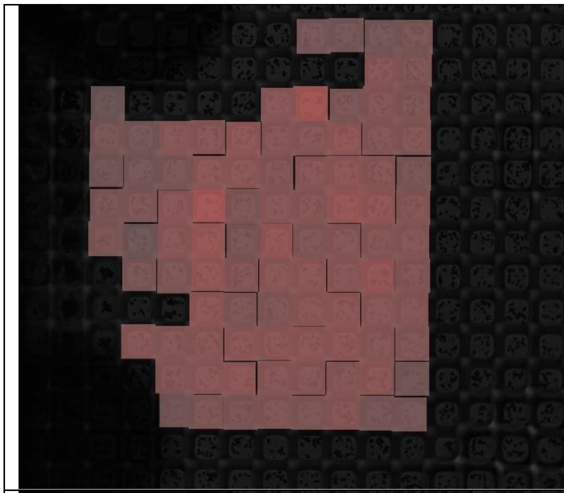
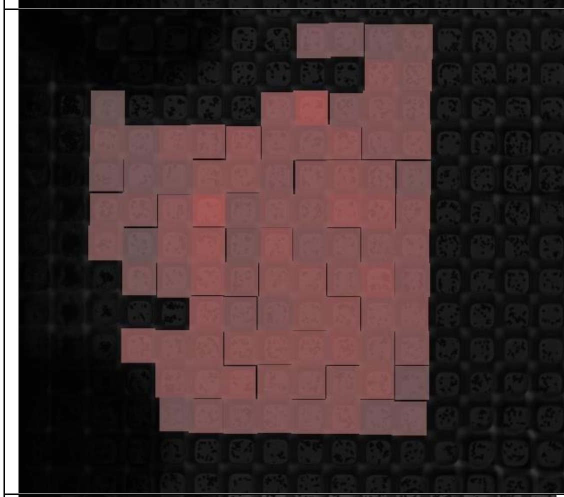
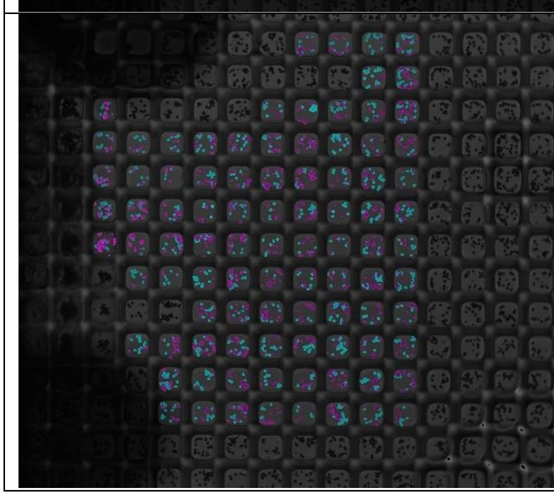
	A	B	C
1	Name	2.8Int	4.5Int
2	0_0_0.jpg	6853.67	2997.212
3	0_0_1.jpg	6393.383	0
4	0_0_10.jpg	6576.141	0
5	0_0_100.jp	6291.371	4883
6	0_0_101.jp	6643.663	2906.5
7	0_0_102.jp	6812.953	0
8	0_0_103.jp	6514.263	0
9	0_0_104.jp	6890.996	0
10	0_0_105.jp	6265.768	0
11	0_0_106.jp	7475.945	0
12	0_0_107.jp	7092.806	0
13	0_0_108.jp	6768.566	3235
14	0_0_109.jp	6804.67	0
15	0_0_11.jpg	6161.192	0
16	0_0_110.jp	7291.741	0
17	0_0_111.jp	6391.709	3081
18	0_0_112.jp	6435.653	0
19	0_0_113.jp	6530.827	3023.7
20	0_0_114.jp	6450.979	3505.611
21	0_0_115.jp	6821.582	0
22	0_0_116.jp	7127.951	0
23	0_0_117.jp	7100.761	7685.913

Figure 12: Example results of prediction

Plotting

Plotting combines data regarding nanowell locations in microscopy images and their respective fluorescent intensities. It also stitches the individual nanowells with normalized intensities to display the intensity distribution in the large microscopy image. To use this function, either calibration or prediction must be conducted. This function will combine FluoroAvgResults.csv and NanowellLocations.csv into CombinedResults.csv. This organizes all the fluorescent measurements and XY coordinates of the cropped nanowell images into a single spreadsheet. Finally, the function uses the coupled data to stitch the individual nanowells with normalized intensities together and overlay with the TIFF images to show the intensity distributions in the microwells. The folder names and the type of stitched images they contain are listed in Table 4.

Images	Folder Name	Description
	Intensity28LOC	2.8um bead intensity boxes using normalization based on the maximum intensity of the TIFF nanowell image
	Intensity28OVR	2.8um bead intensity boxes using normalization based on the maximum intensity of all TIFF nanowell images used

	Intensity45LOC	4.5um bead intensity boxes using normalization based on the maximum intensity of the TIFF nanowell image
	Intensity45OVR	4.5um bead intensity boxes using normalization based on the maximum intensity of all TIFF nanowell images used
	OverallPreds	2.8um and 4.5um bead intensities plotted using the predicted segmentation mask. Normalization based on the maximum intensity of all TIFF nanowell images used

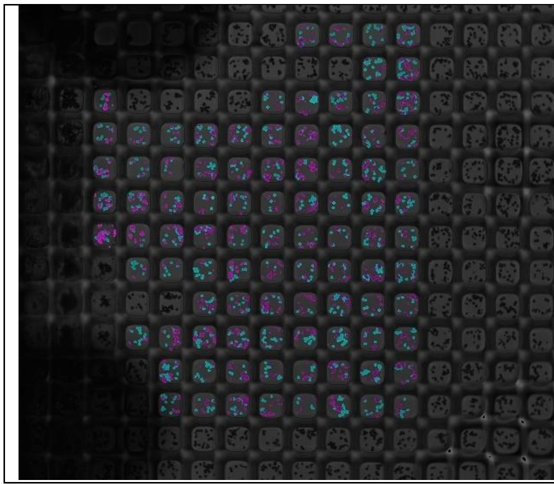
	LocPred	2.8um and 4.5um bead intensities plotted using the predicted segmentation mask. Normalization based on the maximum intensity of the TIFF nanowell image

Table 4: Stitched Nanowell intensity plots.4.5um beads will generally show as blue in the Overall Preds and LocPred images, 2.8um beads will show as purple.

Final Remarks

All results regarding the fluorescent intensity measurements from calibrations and predictions are on a 16-bit pixel scale. Therefore, the highest intensity value of a pixel is 2^{16}-1. The average pixel intensity is calculated for each nanowell in both the calibration and prediction sections.

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